

Algebra II with Trigonometry

Chapter 5.2

Olympiad Problems (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:

[4401268_alg-2trig-h-chp-5-2-note-docx.pdf](#)

Problems

1. An observer looking at the top of a radio tower measures an angle of elevation of 15° . After walking exactly 100 meters directly toward the tower on level ground, the angle of elevation increases to 30° . Mentally compute the exact height of the tower in meters.
2. Without using a calculator, determine the exact numerical value of the expression: $(\sin 15^\circ + \cos 15^\circ)^2$
3. Given that $\sec \theta + \tan \theta = 5$ for an acute angle θ , mentally evaluate the exact value of $\sec \theta - \tan \theta$.
4. Exploit the symmetries of the right triangle to evaluate the following sum entirely in your head:
 $\cos^2 10^\circ + \cos^2 20^\circ + \cos^2 30^\circ + \cos^2 40^\circ + \cos^2 50^\circ + \cos^2 60^\circ + \cos^2 70^\circ + \cos^2 80^\circ$
5. Using the definitions of the trigonometric ratios, find the exact value of the following 89-term product: $\tan 1^\circ \cdot \tan 2^\circ \cdot \tan 3^\circ \cdots \tan 88^\circ \cdot \tan 89^\circ$
6. In a right triangle with acute angles α and β , it is known that $\log_2(\sin \alpha) + \log_2(\cos \alpha) = -2$. Mentally determine the degree measures of both α and β .